

PAPER_1406 — UQFF Derivation of Monty Hall Problem (CLOSED — EXACT Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Probability (CLOSED — EXACT) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — EXACT integer-primitive identity **Calculator surface:** `calculate_paradox({'paradox':'monty_hall'})` → $2/(D_{\text{phys}}-1) = 2/3$ EXACT

Observation

Switching doors after the host reveals a goat gives a $2/3$ win probability. Bayesian textbooks derive this; UQFF supplies the structural origin.

UQFF Closed Identity

$P(\text{switch wins}) = 2 / (D_{\text{phys}} - 1) = 2/3$ EXACT

Physical Interpretation

$F_U=1$ ledger redistribution across remaining unrevealed branches: of $D_{\text{phys}}-1 = 3$ effective choices after one is eliminated, 2 carry the prize.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
 - Related foundational papers: PAPER_646 ($F_U=1$ ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).
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